

GENX: Porewater Methane

May 2026 Samples

2026-06-11

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##Run Information

```
cat("Run Information: Melanie Giessner ") #lets you know what section you're in
```

Run Information: Melanie Giessner

```
#set the run date & user name
sample_year <- "2026"
sample_month <- "May"
user <- "Melanie Giessner"
#identify the files you want to read in
#read in as a list to accommodate multiple runs in a month

#change dates
folder_path_rawdata <- paste0("Raw Data")
GHG_files <- list.files(path = folder_path_rawdata,pattern = "20260513" ,full.names = TRUE)
head(GHG_files)
```

```
## [1] "Raw Data/20260513_GENX_GHGdata.csv"
```

```
#GHG_files <- c("Raw Data/20260417_GENX_GHGdata.csv","Raw Data/20260501_GENX_GHGdata.csv" )
```

```
# Define the file path for QAQC log file - NO Need to change just check year
log_path <- "Processed Data/ShimadzuGC_Slope_QAQC.csv"
```

```
# Define final path for data be sure to change year and month and run number
final_path <- "Processed Data/GCReW_GENX_Porewater_CH4_20260513.csv"
```

```
#Gas in Water Calculations
# "TempC" column value in Celsius (equilibration temperature) and volume of gas (in liters) in syringe
TempC = 25 #Equilibration Temperature (Celsius)
```

```
volume_g = 0.015 #volume of gas (in liters) in syringe when equilibrated
volume_w = 0.015 #volume of water (in liters) in syringe when equilibrated (should be equal to volume_g)
```

```
#record any notes about the run or anything other info here:
run_notes <- ""
```

```
#Metadata File Path
Raw_Metadata = here("..", "GCReW", "GCReW_Project_Treatment_Metadata.csv")
```

```
#Collection_Dates = "Raw Data/Sample_Collection_Dates_2024.csv"
```

```
cat(run_notes)
```

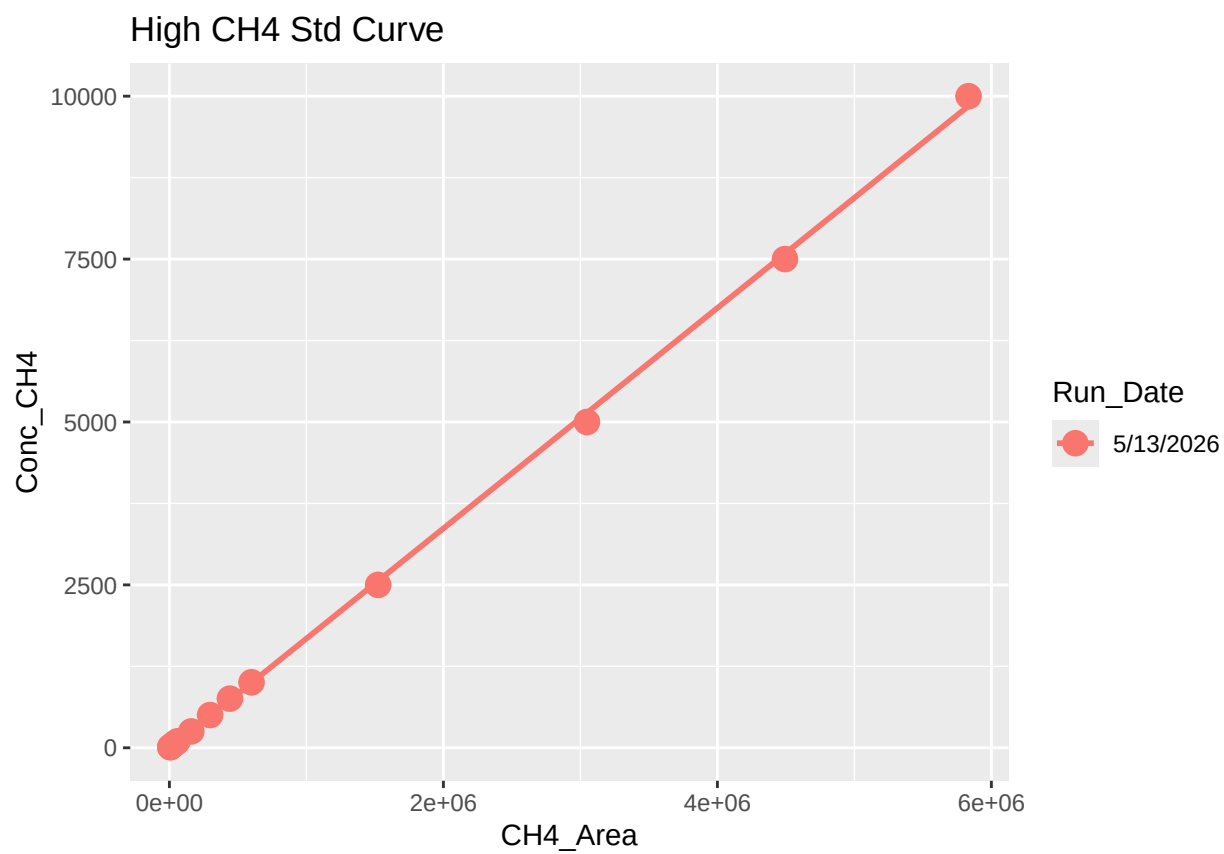
##Set Up

##Read in metadata and create similar sample IDs for matching to samples

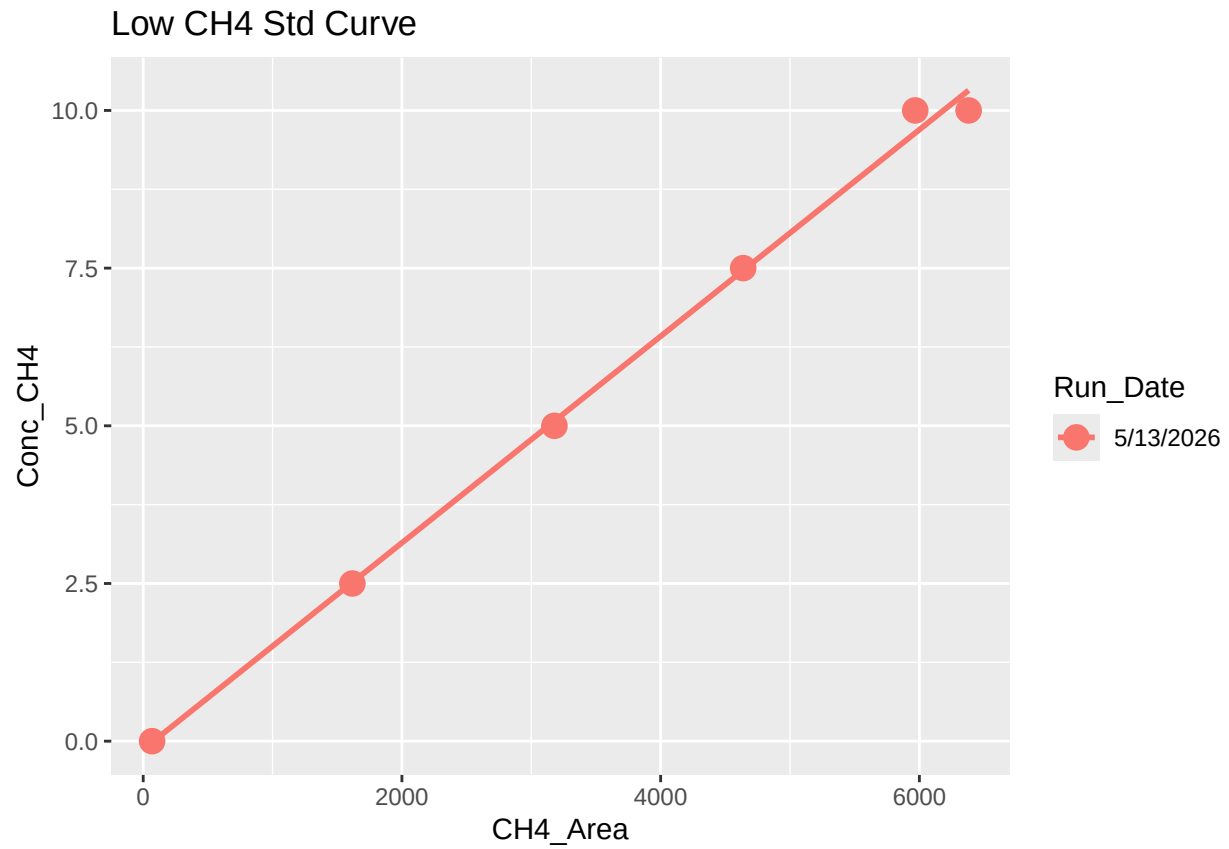
1 Read in data files

2 Assess standard curves

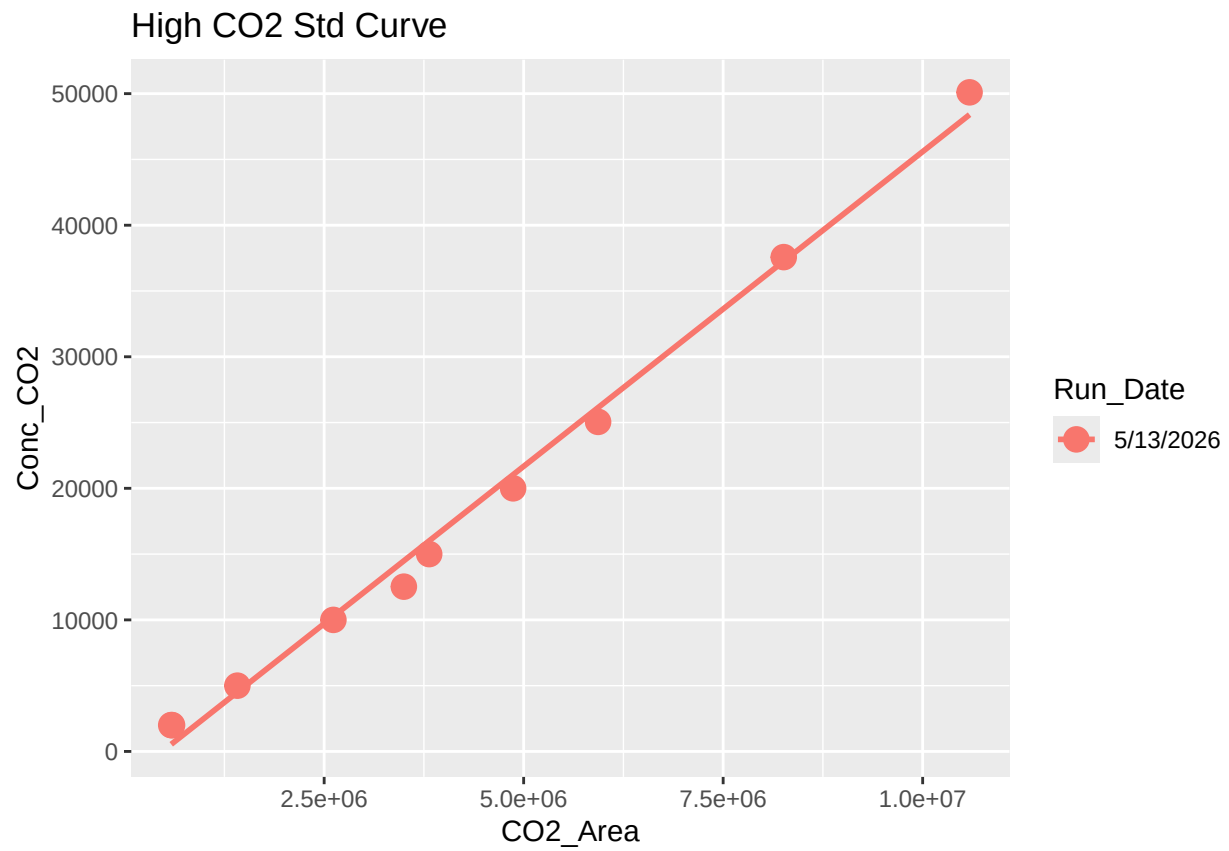
```
## 'geom_smooth()' using formula = 'y ~ x'
```



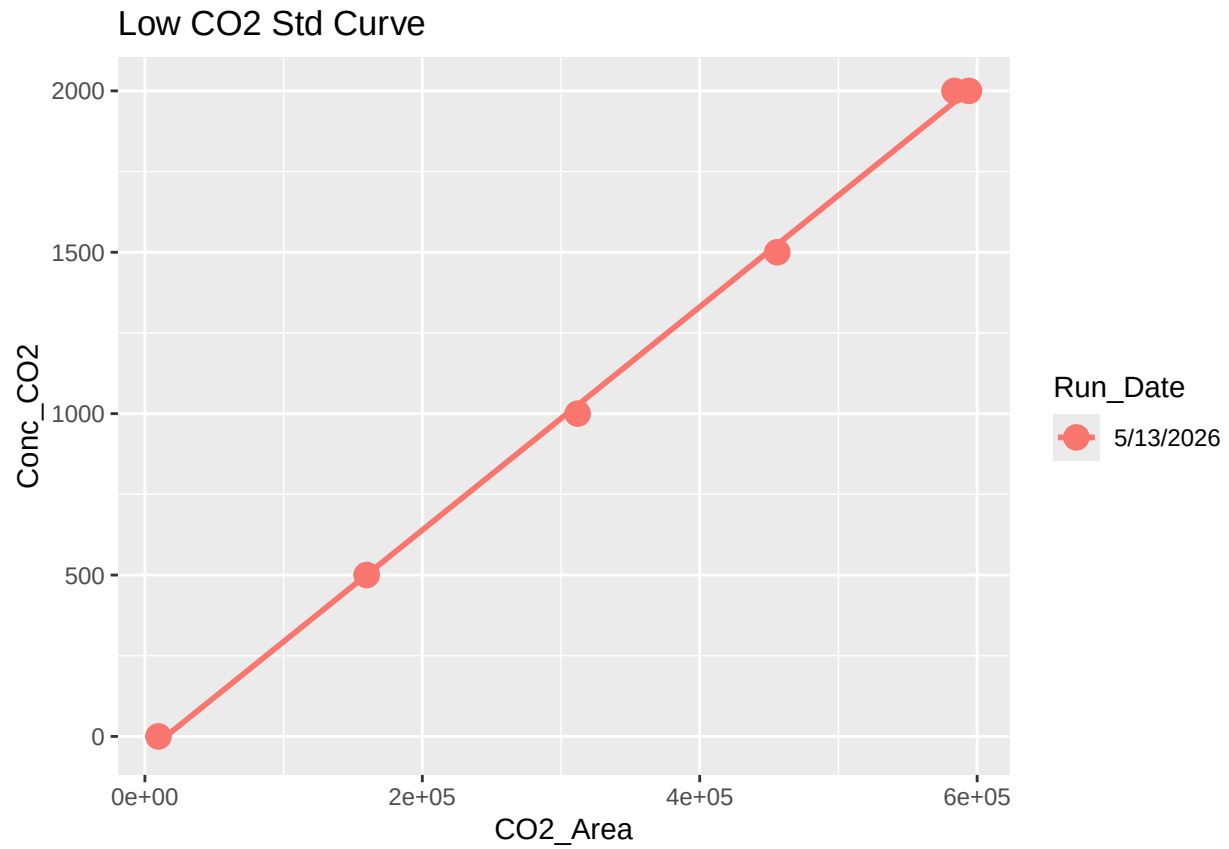
```
## 'geom_smooth()' using formula = 'y ~ x'
```



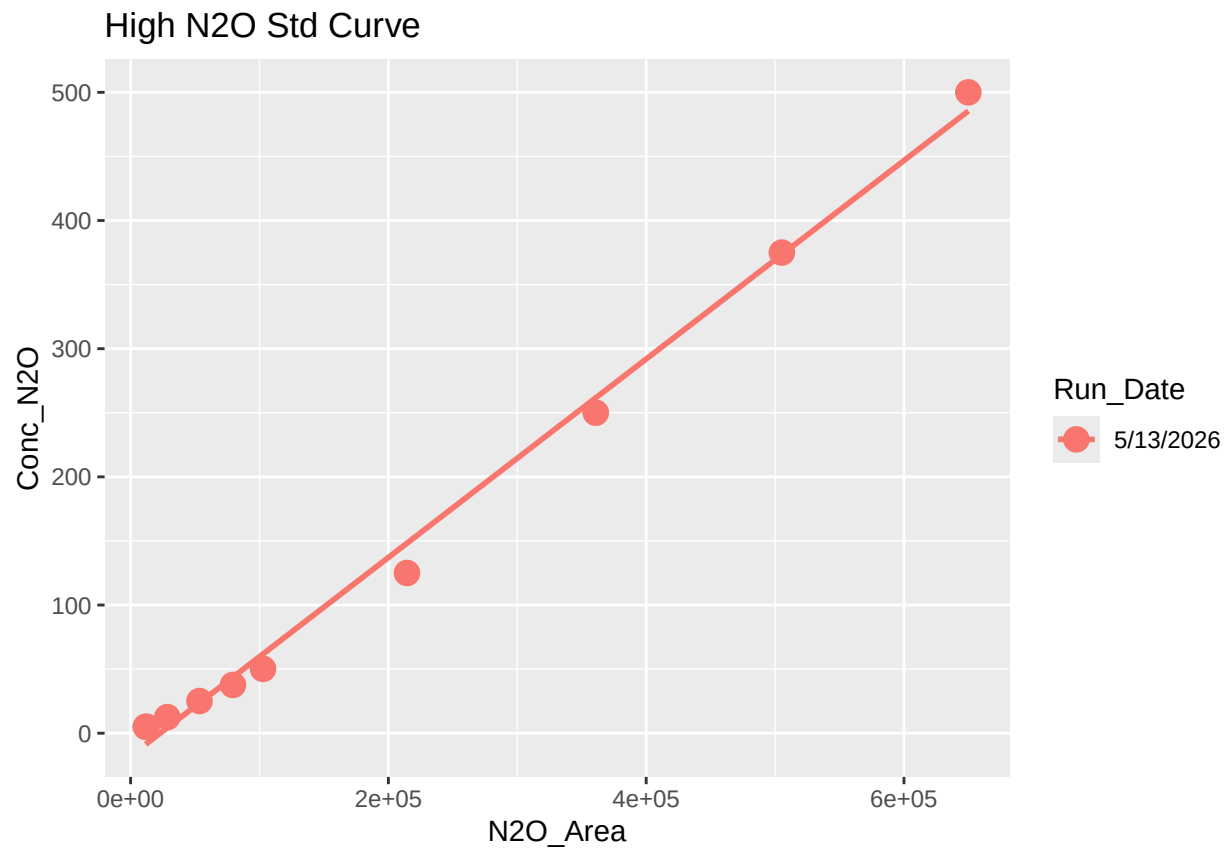
```
## Joining with 'by = join_by(Run_Date)'  
  
## [1] "High_R2 is Good-5/13/2026"  
  
## [1] "Low_R2 is Good-5/13/2026"  
  
## 'geom_smooth()' using formula = 'y ~ x'
```



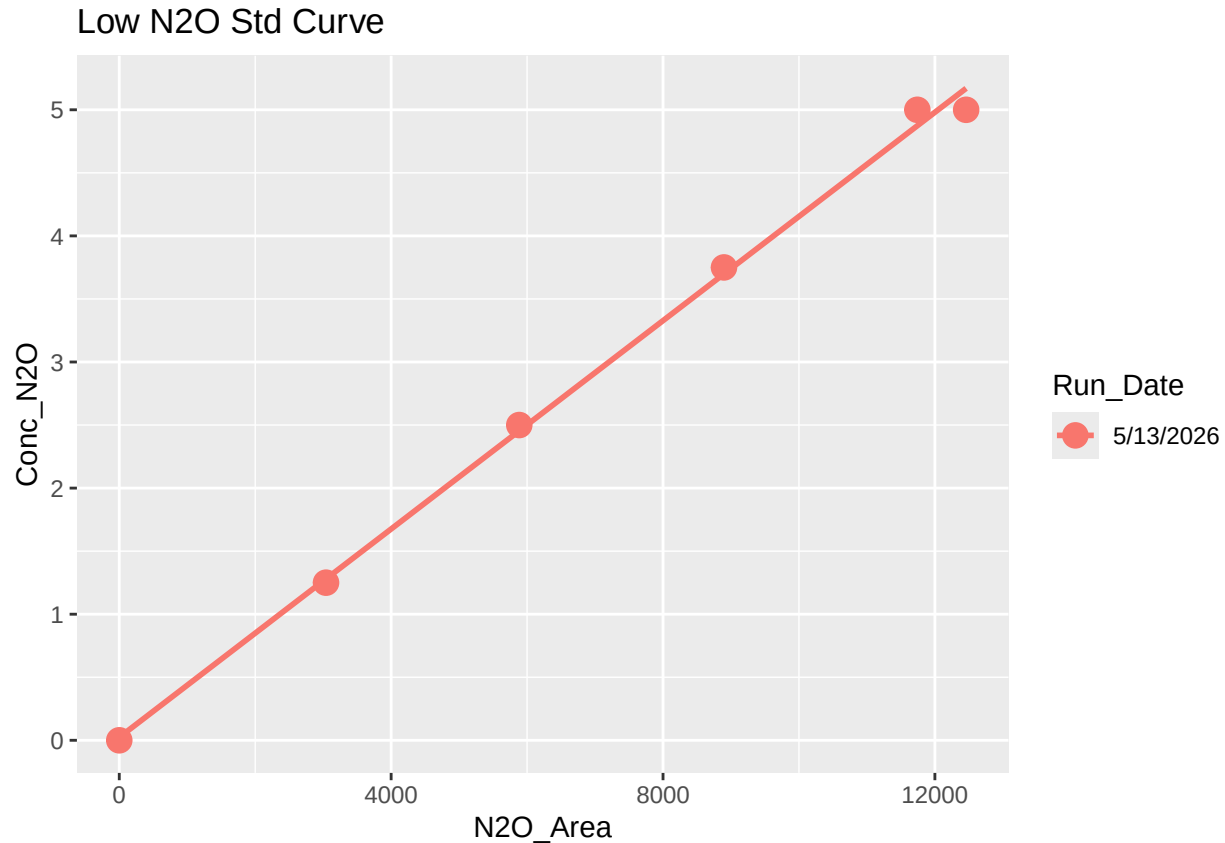
```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
## Joining with 'by = join_by(Run_Date)'  
  
## [1] "High_R2 is Good-5/13/2026"  
  
## [1] "Low_R2 is Good-5/13/2026"  
  
## 'geom_smooth()' using formula = 'y ~ x'
```



```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
## Joining with 'by = join_by(Run_Date)'
```

```
## [1] "High_R2 is Good-5/13/2026"
```

```
## [1] "Low_R2 is Good-5/13/2026"
```

```
#add new standard curve data to log
```

```
Slope_CH4_1 <- Slope_CH4 %>% mutate(curve = paste0("Slope_CH4_", CH4_Curve), R2 = CH4_R2, slope = CH4_S
```

```
  CH4_Curve == "Low" ~ paste(CH4_low_std_crv_cutoff,"ppm"),
```

```
  CH4_Curve == "High" ~ paste(CH4_high_std_crv_cutoff,"ppm"),
```

```
  .default = as.character(CH4_mid_std_crv_cutoff)
```

```
) %>% dplyr::select(Analysis, Run_Date, Run_by, Curve_top, curve, R2, slope, intercept)
```

```
Slope_CO2_1 <- Slope_CO2 %>% mutate(curve = paste0("Slope_CO2_", CO2_Curve), R2 = CO2_R2, slope = CO2_S
```

```
  CO2_Curve == "Low" ~ paste(CO2_low_std_crv_cutoff,"ppm"),
```

```
  CO2_Curve == "High" ~ paste(CO2_high_std_crv_cutoff,"ppm"),
```

```
  .default = as.character(CO2_mid_std_crv_cutoff)
```

```
) %>% dplyr::select(Analysis, Run_Date, Run_by, Curve_top, curve, R2, slope, intercept)
```

```
Slope_N2O_1 <- Slope_N2O %>% mutate(curve = paste0("Slope_N2O_", N2O_Curve), R2 = N2O_R2, slope = N2O_S
```

```
  N2O_Curve == "Low" ~ paste(N2O_low_std_crv_cutoff,"ppm"),
```

```
  N2O_Curve == "High" ~ paste(N2O_high_std_crv_cutoff,"ppm"),
```

```
  .default = as.character(N2O_mid_std_crv_cutoff)
```

```
) %>% dplyr::select(Analysis, Run_Date, Run_by, Curve_top, curve, R2, slope, intercept)
```

```
Slope <- rbind(Slope_CH4_1, Slope_CO2_1, Slope_N2O_1)
```

```
log <- read.csv(log_path)
log <- log[, -c(1)]
head(log)
```

```
##   Analysis Run_Date      Run_by Curve_top      curve      slope
## 1    CH4 7/9/2024 Alia Al-Haj    10 ppm Slope_Ch4_low 0.001594508
## 2    CH4 7/9/2024 Alia Al-Haj   100 ppm Slope_Ch4_high 0.001562326
## 3    N2O 7/9/2024 Alia Al-Haj    5 ppm Slope_N2O_low 0.000192023
## 4    N2O 7/9/2024 Alia Al-Haj  50.2 ppm Slope_N2O_high 0.000282245
## 5    CO2 7/9/2024 Alia Al-Haj  2000 ppm Slope_CO2_low 0.000282245
## 6    CO2 7/9/2024 Alia Al-Haj 20000 ppm Slope_CO2_high 0.001549502
##      intercept      R2
## 1 -0.20901179 0.9960215
## 2  0.09646243 0.9996531
## 3  0.07848462 0.9940785
## 4 -0.49925490 0.9960644
## 5 -0.49925490      NA
## 6 38.64567509 0.9996907
```

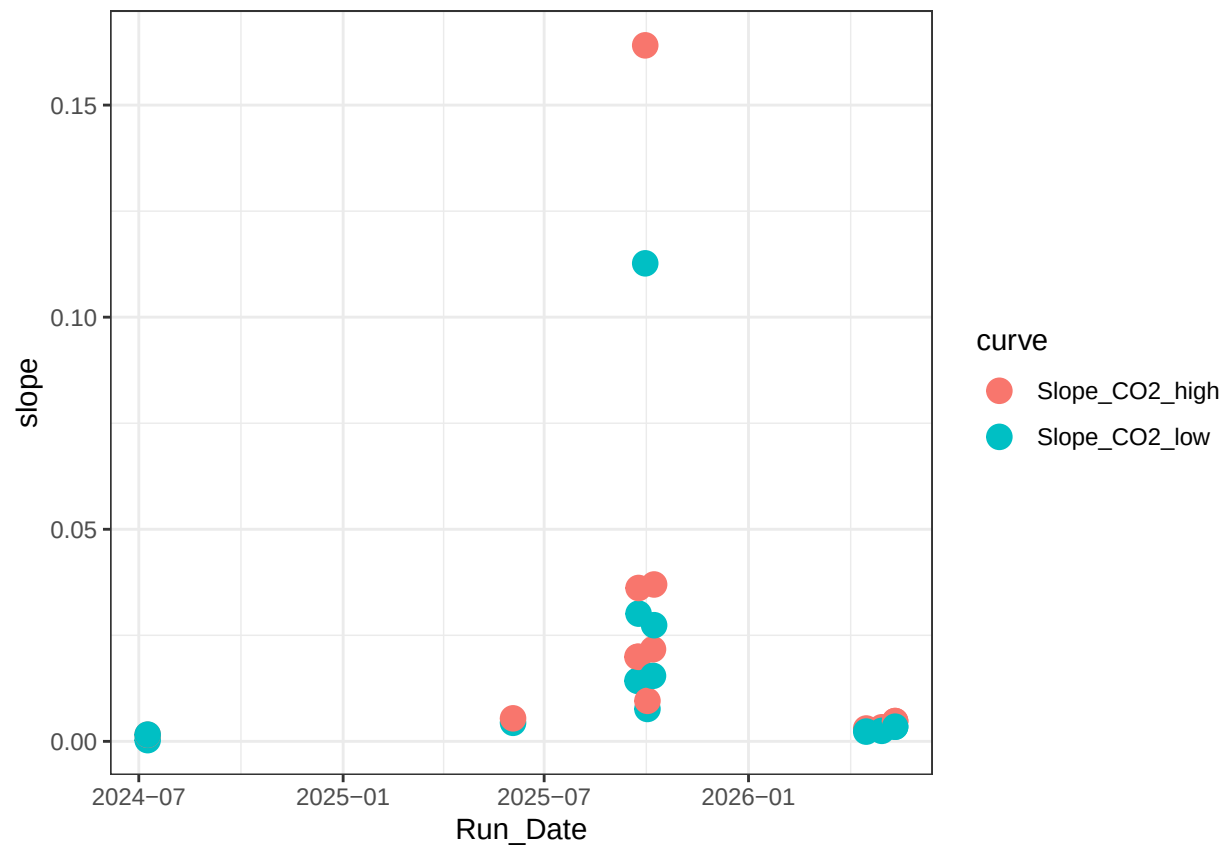
```
log <- rbind(log, Slope)
```

#remove possible duplicates from multiple QAQC attempts

```
unique_log <- log %>%
  mutate(Run_Date = as.Date(Run_Date, tryFormats = c("%Y-%m-%d", "%m/%d/%Y", "%m/%d/%y")), curve= gs)

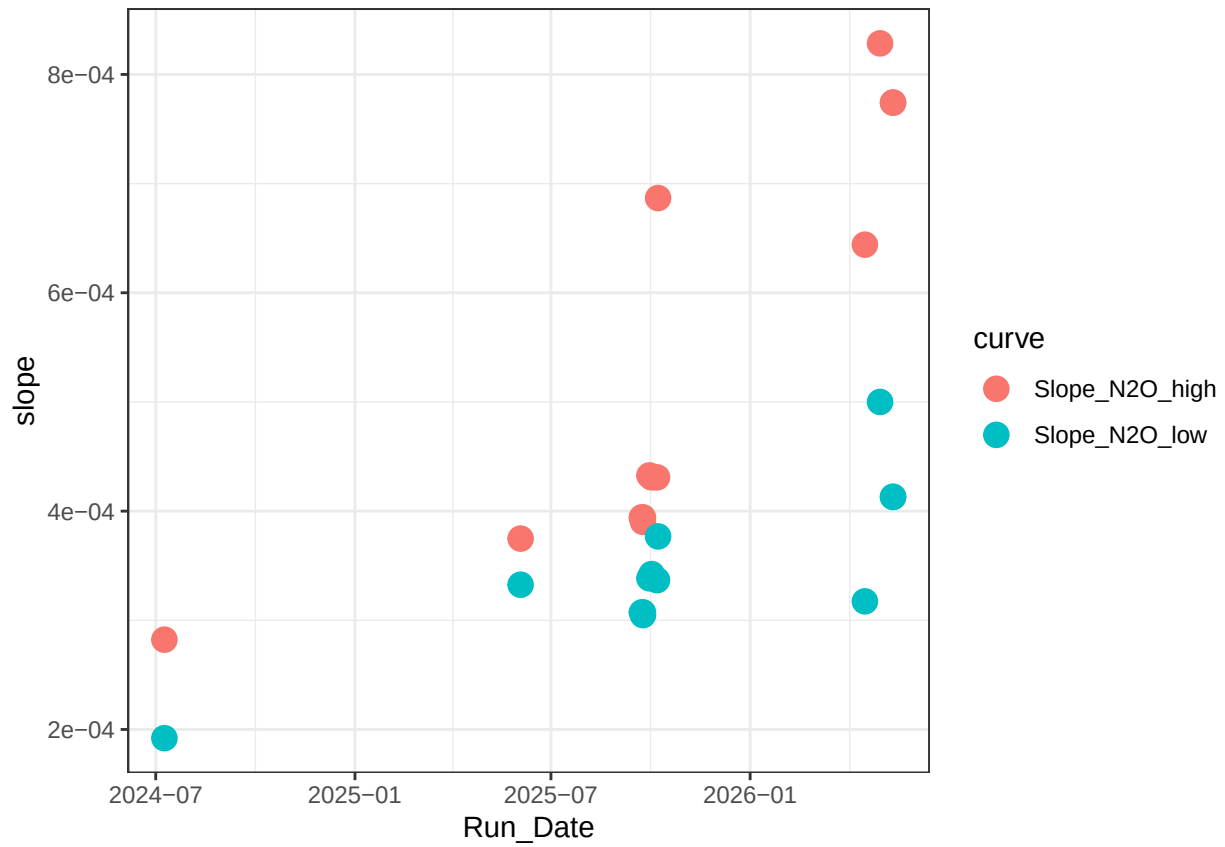
Slopes_chk_CH4 <- ggplot(unique_log %>% filter( grepl("CH4", curve, ignore.case = T)), aes(x = Run_Date,
  geom_point(size = 4) +
  #ylim(400, 800) +
  theme_bw())
```

```
Slopes_chk_CH4
```

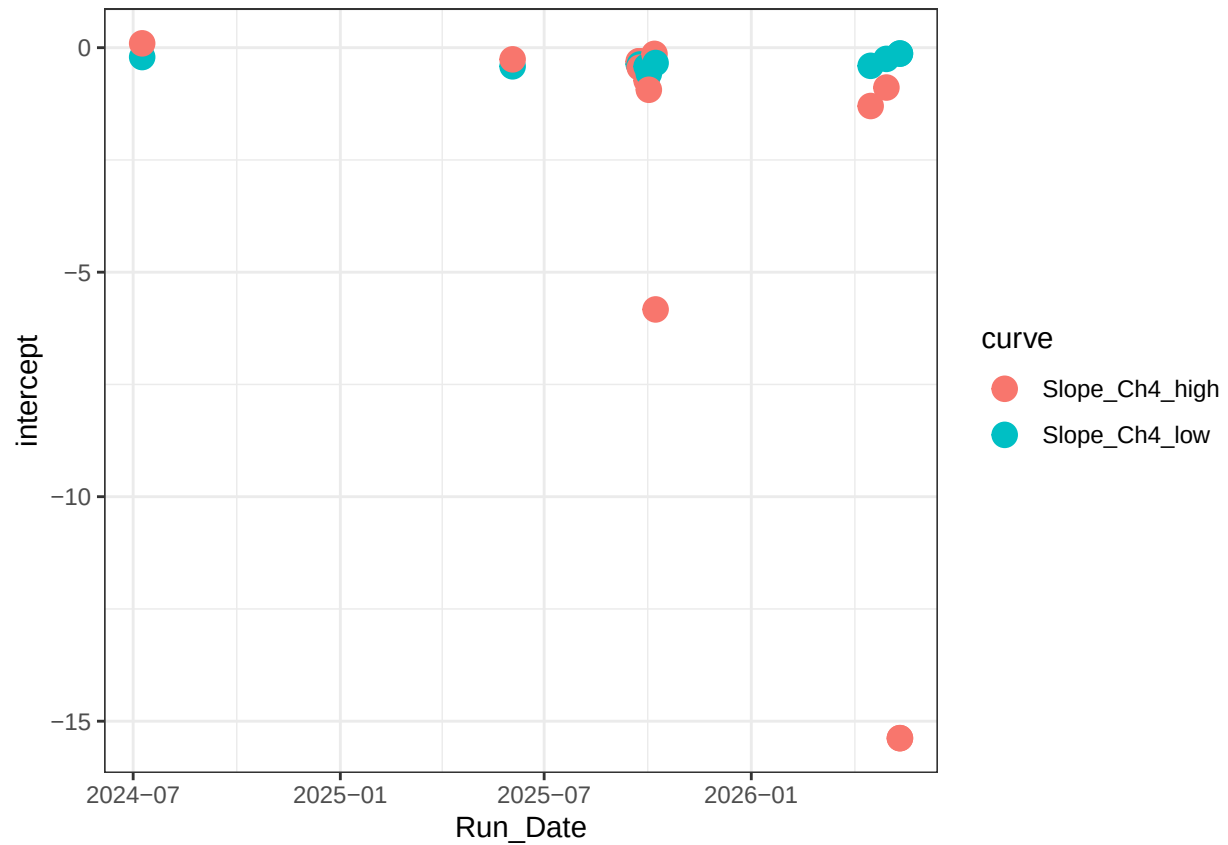
```
Slopes_chk_N20 <- ggplot(unique_log %>% filter( grepl("N20",curve,ignore.case = T)), aes(x = Run_Date,
  geom_point(size = 4) +
  #ylim(400, 800) +
  theme_bw()
```

Slopes_chk_N20



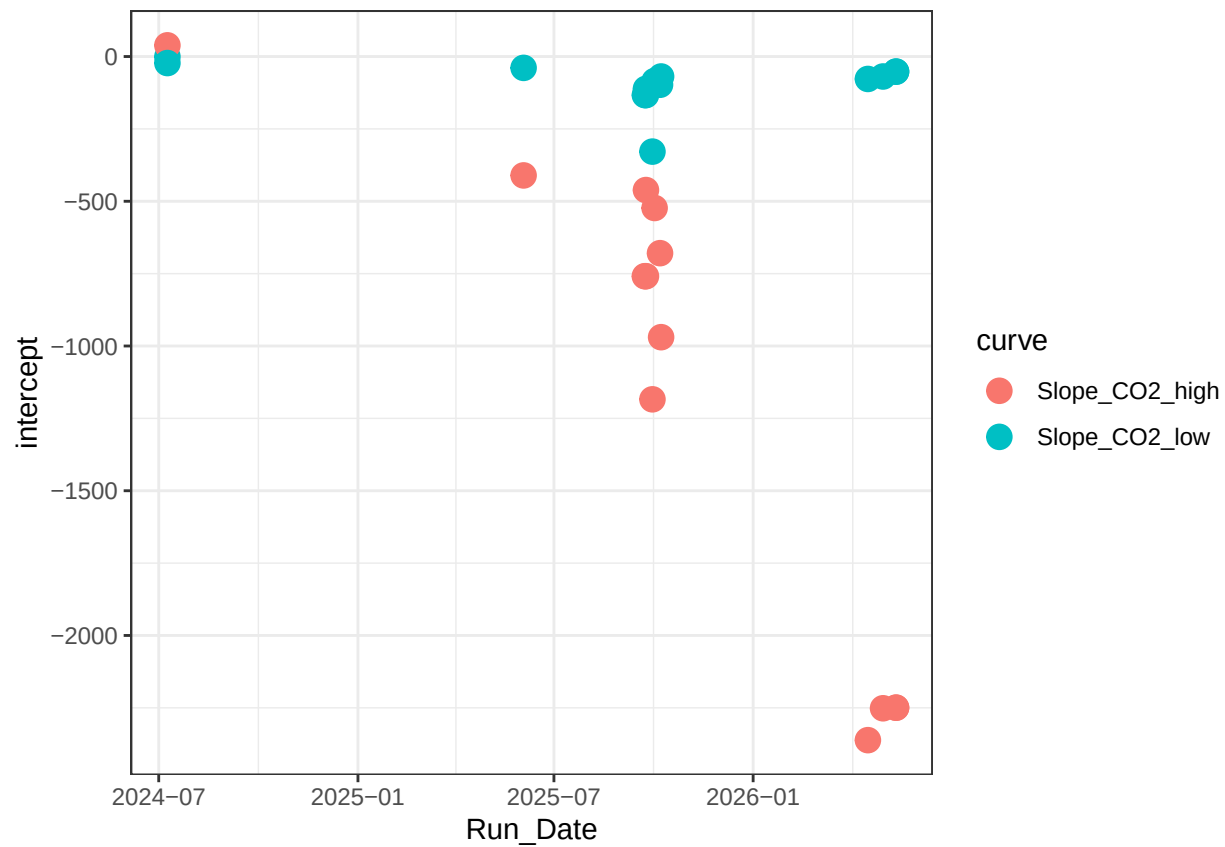
```
int_chk_CH4 <- ggplot(unique_log %>% filter( grepl("CH4",curve,ignore.case = T)), aes(x = Run_Date, y = slope)) +
  geom_point(size = 4) +
  #ylim(400, 800) +
  theme_bw()

int_chk_CH4
```



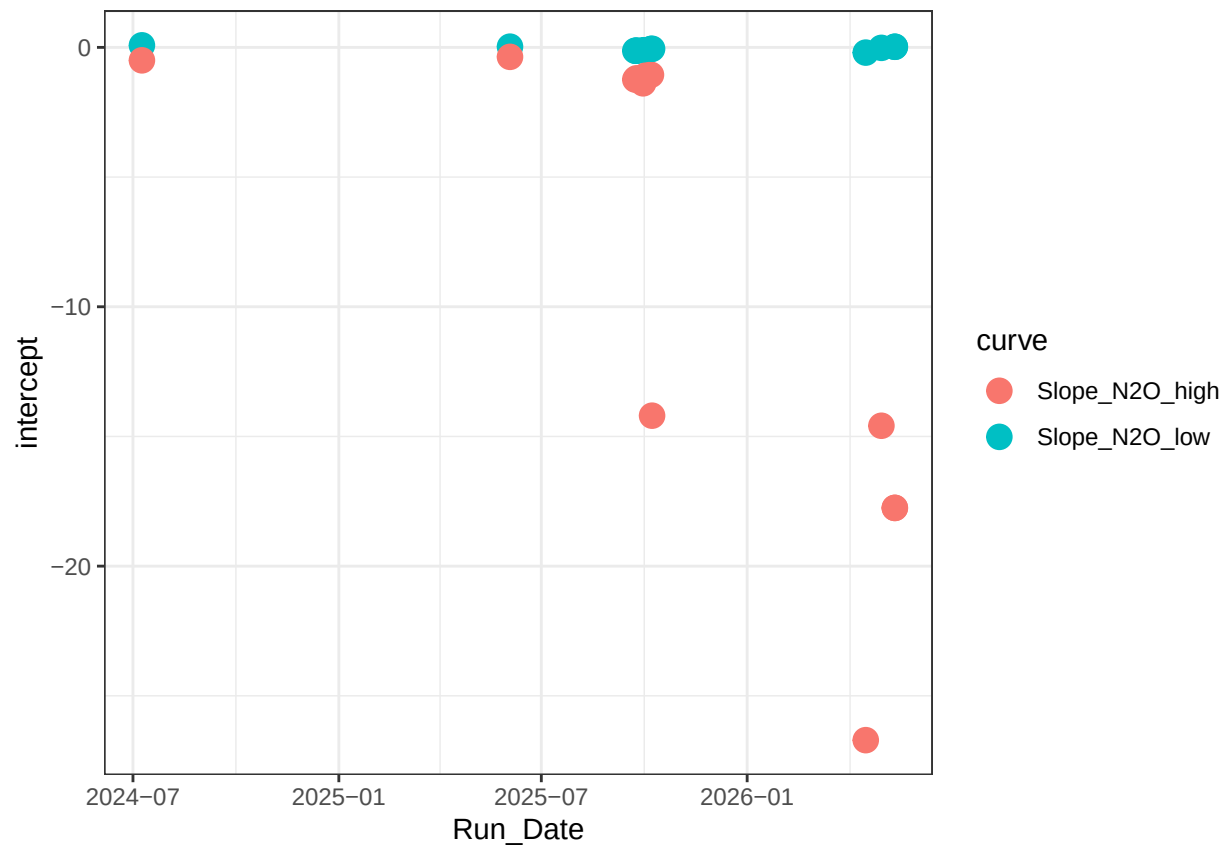
```
int_chk_C02 <- ggplot(unique_log %>% filter( grepl("C02",curve,ignore.case = T)), aes(x = Run_Date, y = intercept)) +
  geom_point(size = 4) +
  #ylim(400, 800) +
  theme_bw()

int_chk_C02
```



```
int_chk_N20 <- ggplot(unique_log %>% filter( grepl("N20",curve,ignore.case = T)), aes(x = Run_Date, y = intercept)) +
  geom_point(size = 4) +
  #ylim(400, 800) +
  theme_bw()
```

```
int_chk_N20
```



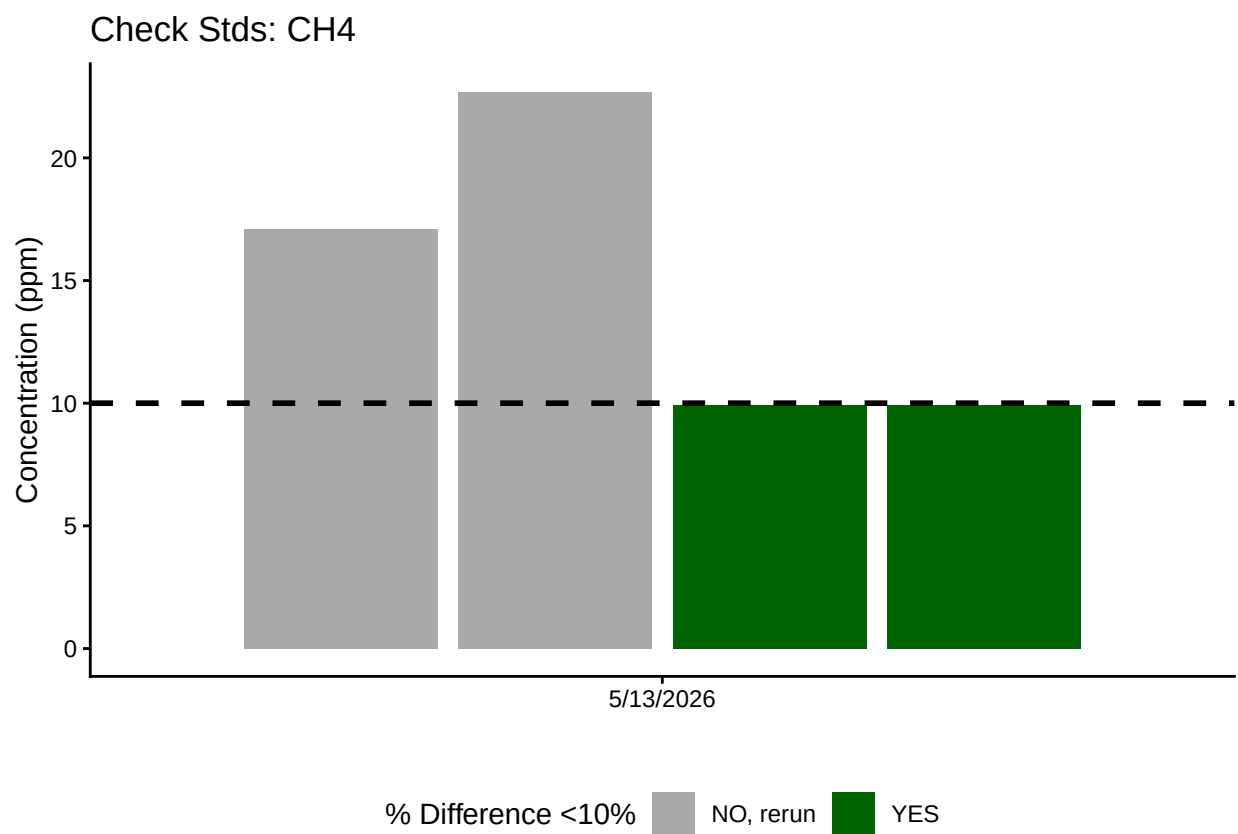
2.1 Now calculate the CH₄ in ppm

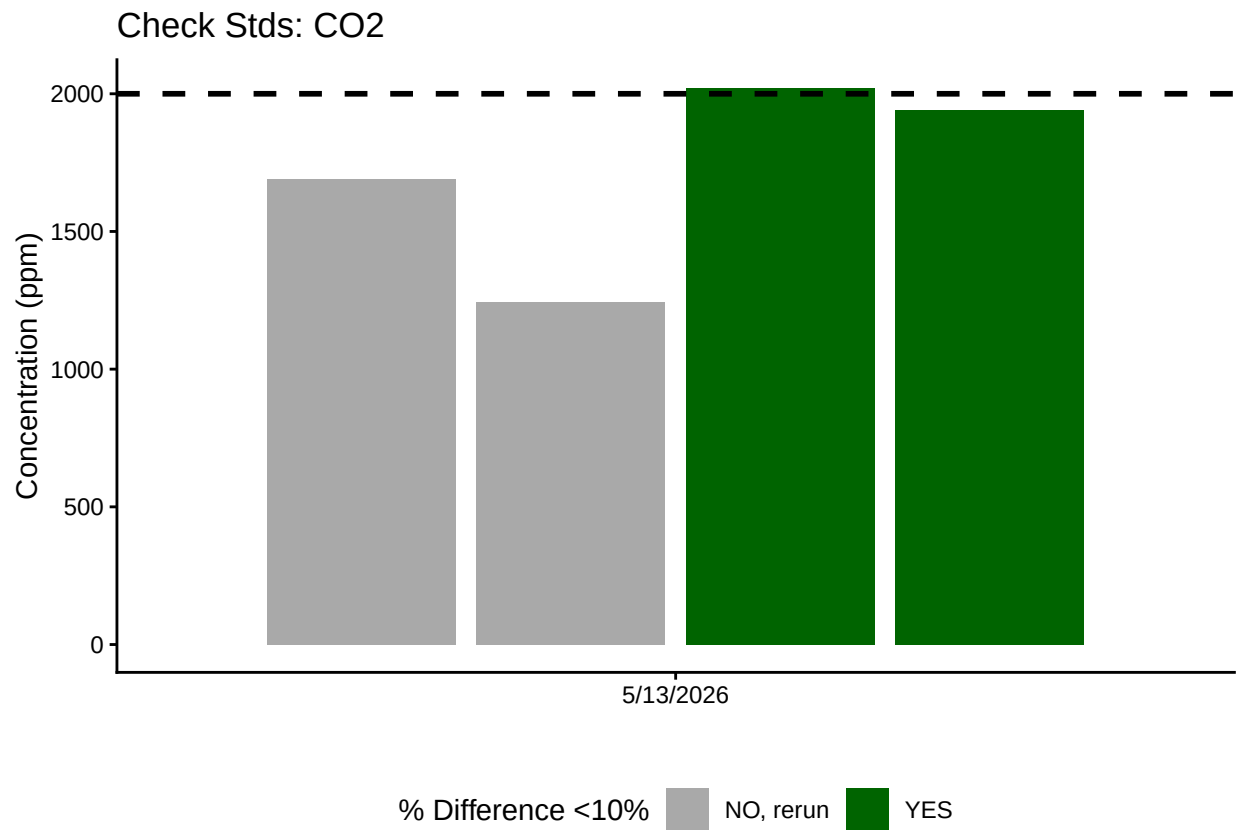
3 Aanlayze the Check Standards

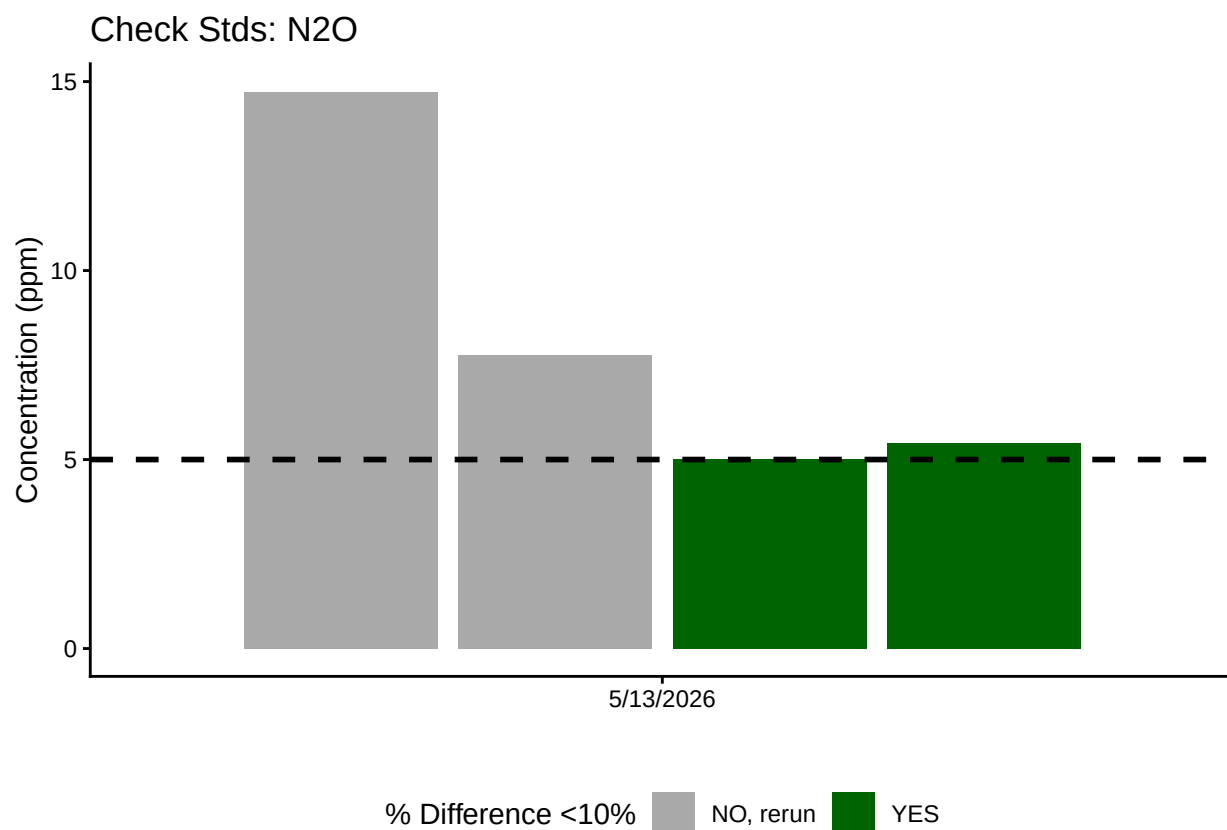
<60% of CH₄ Check Standards are within range of expected concentration - REASSESS

<60% of CO₂ Check Standards are within range of expected concentration - REASSESS

<60% of N₂O Check Standards are within range of expected concentration - REASSESS







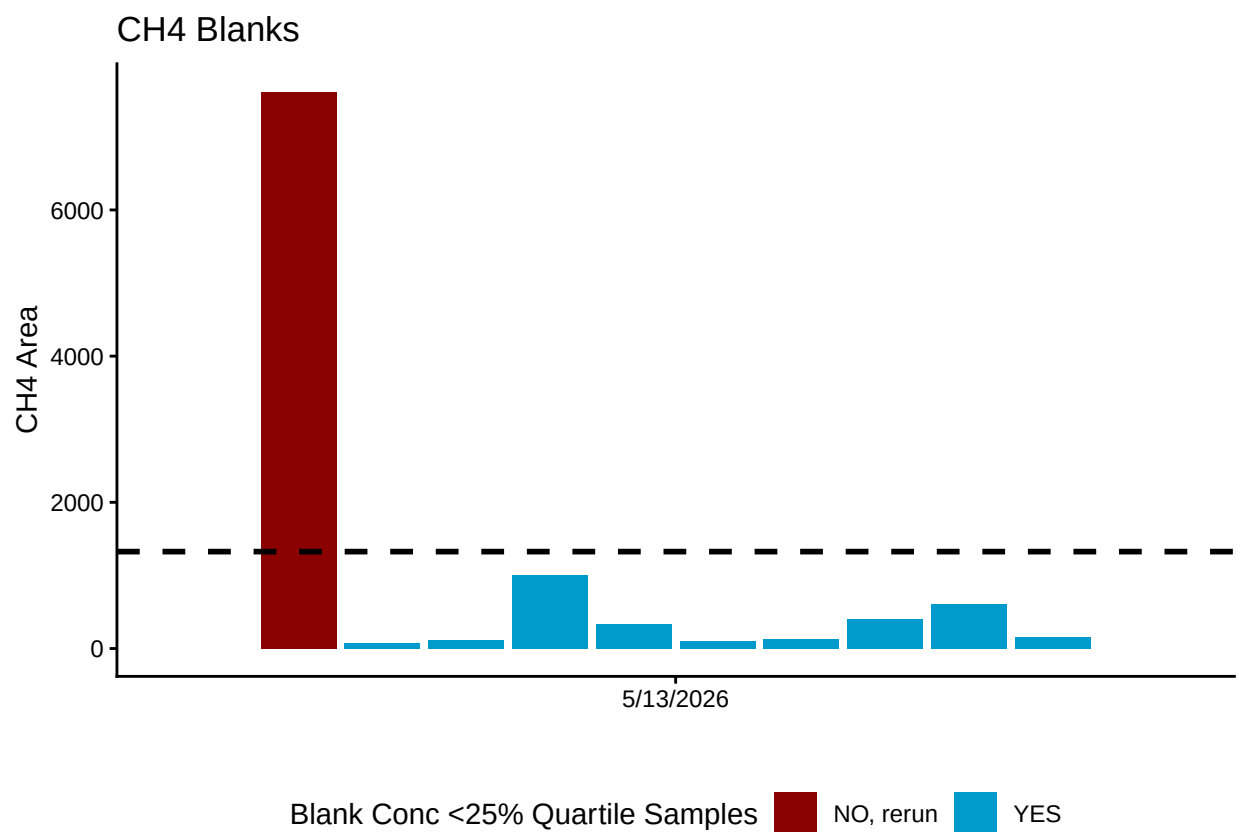
4 Analyze the Blanks

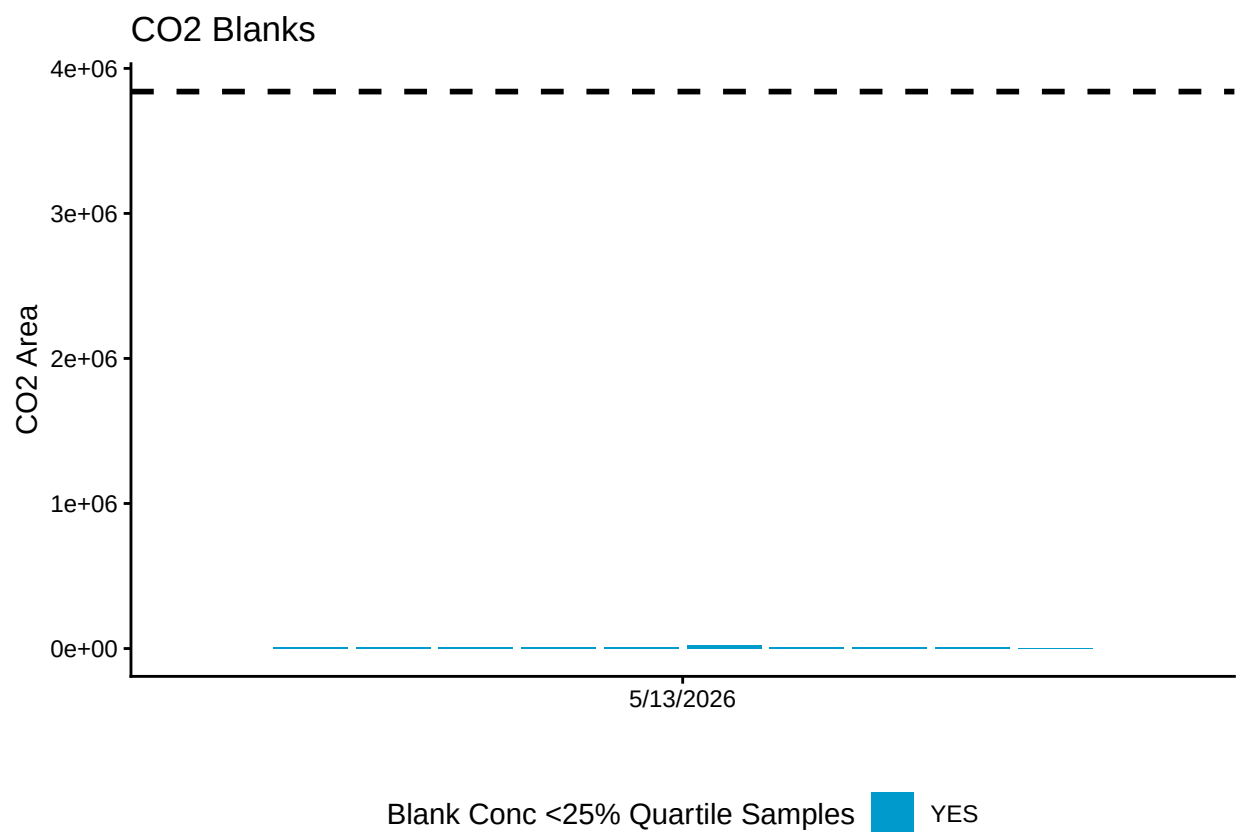
Assess Blanks

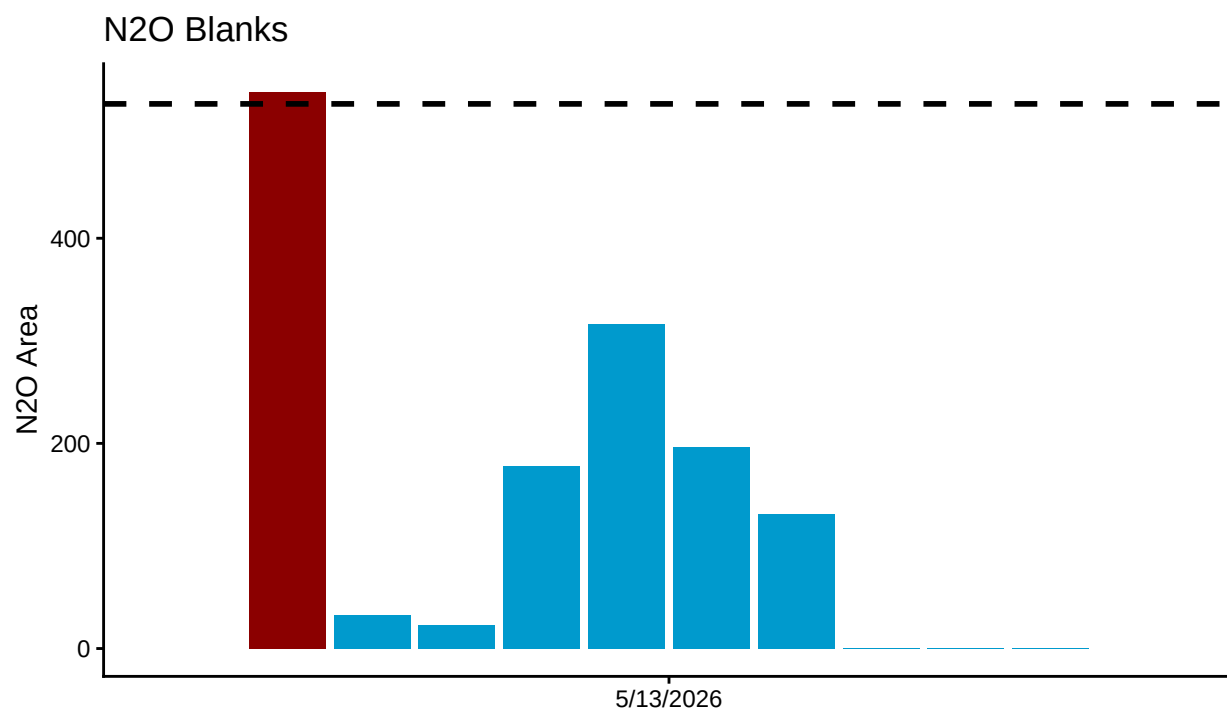
>60% of CH₄ Blanks are below the lower 25% quartile of samples - PROCEED

>60% of CO₂ Blanks are below the lower 25% quartile of samples - PROCEED

>60% of N₂O Blanks are below the lower 25% quartile of samples - PROCEED





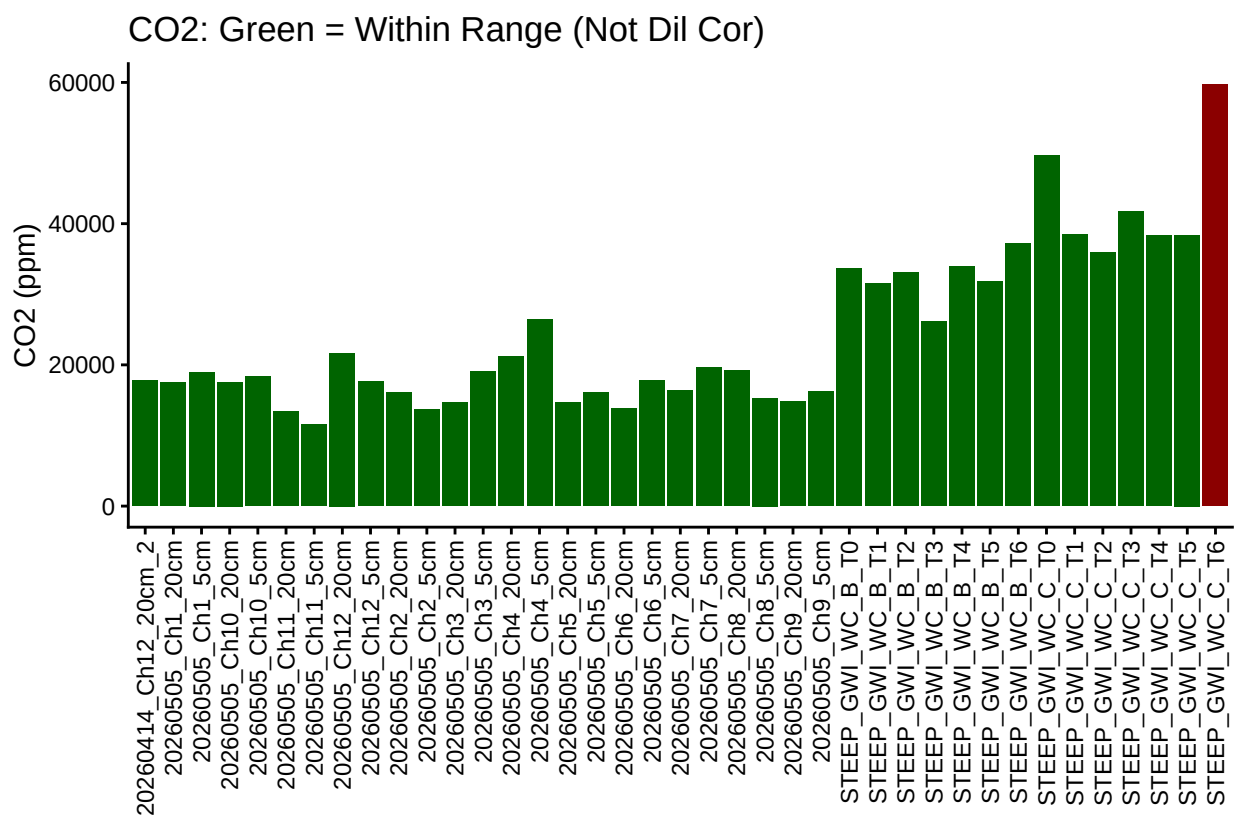
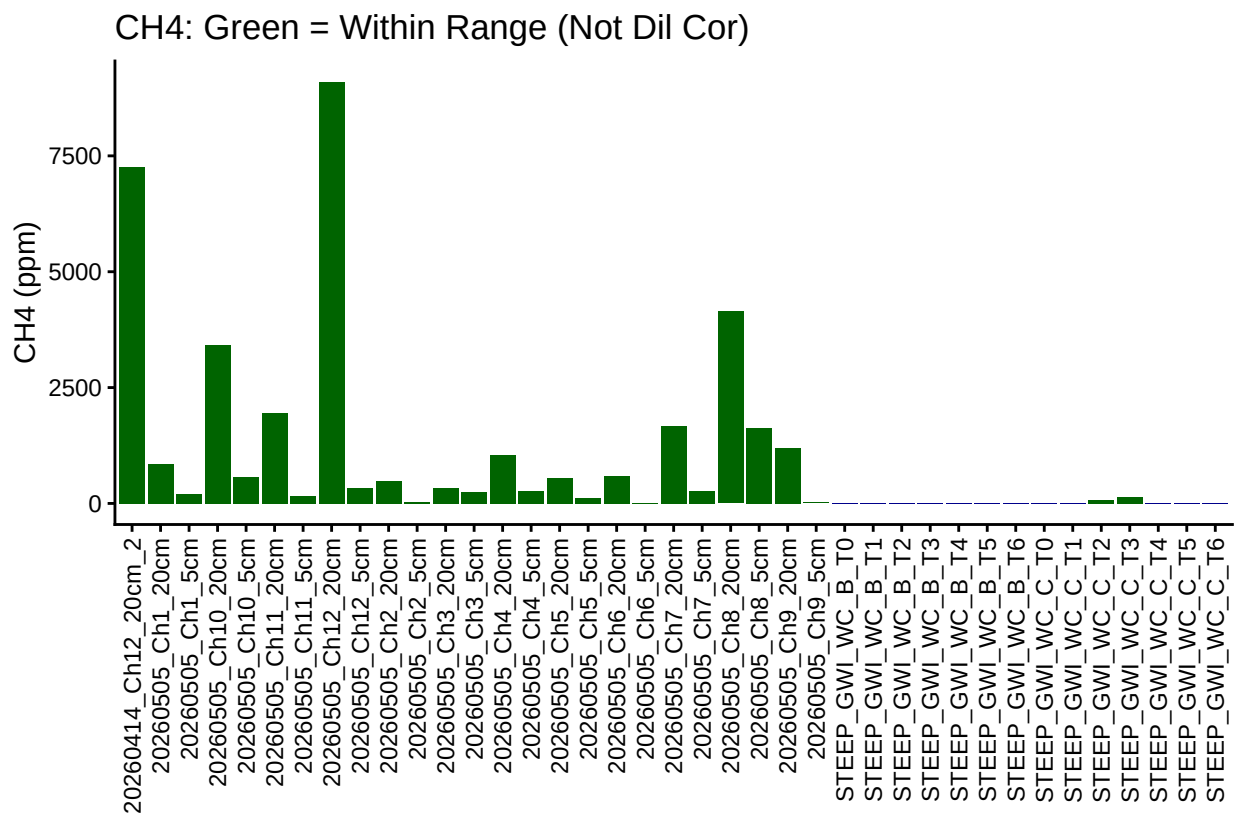


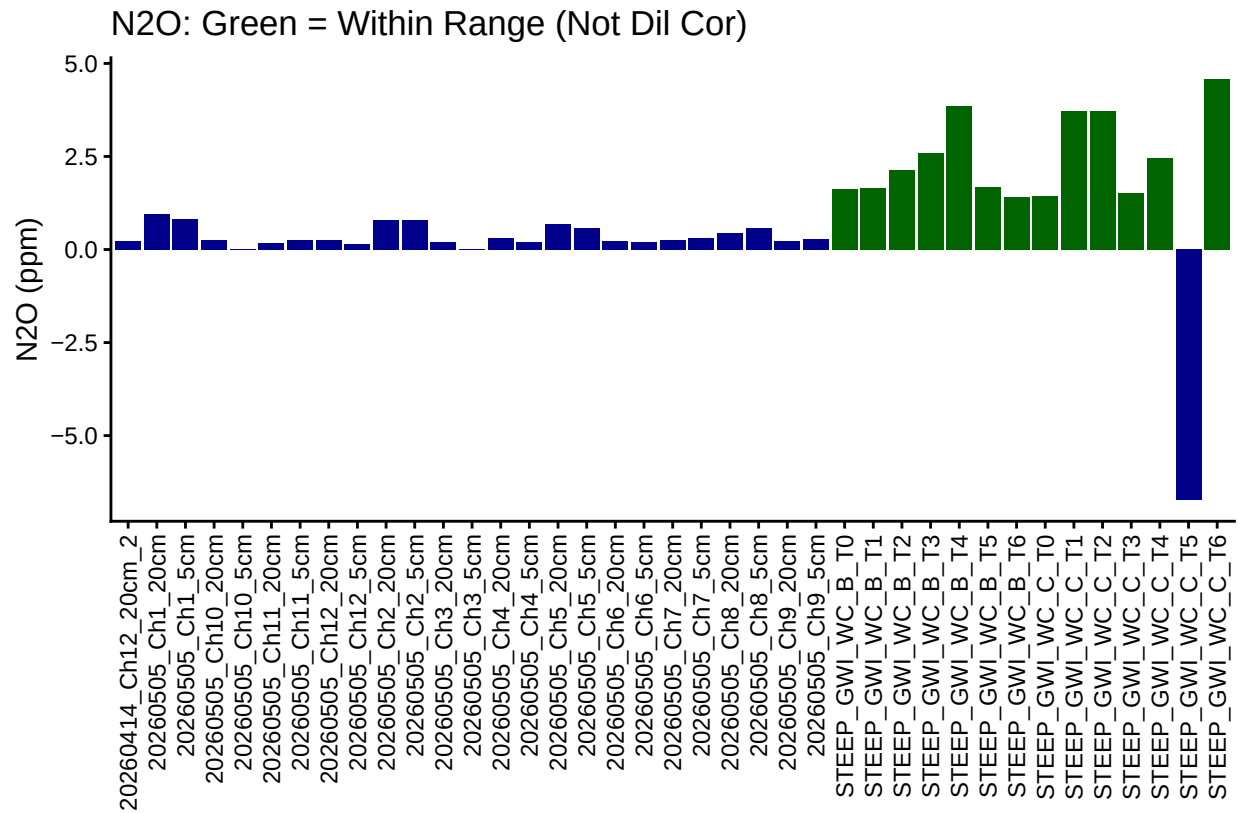
Blank Conc <25% Quartile Samples ■ NO, rerun ■ YES

Table 1: Mean Area of Blanks

Test	Blank_Mean_Area
CH4	1055.8
CO2	11286.6
N2O	142.0

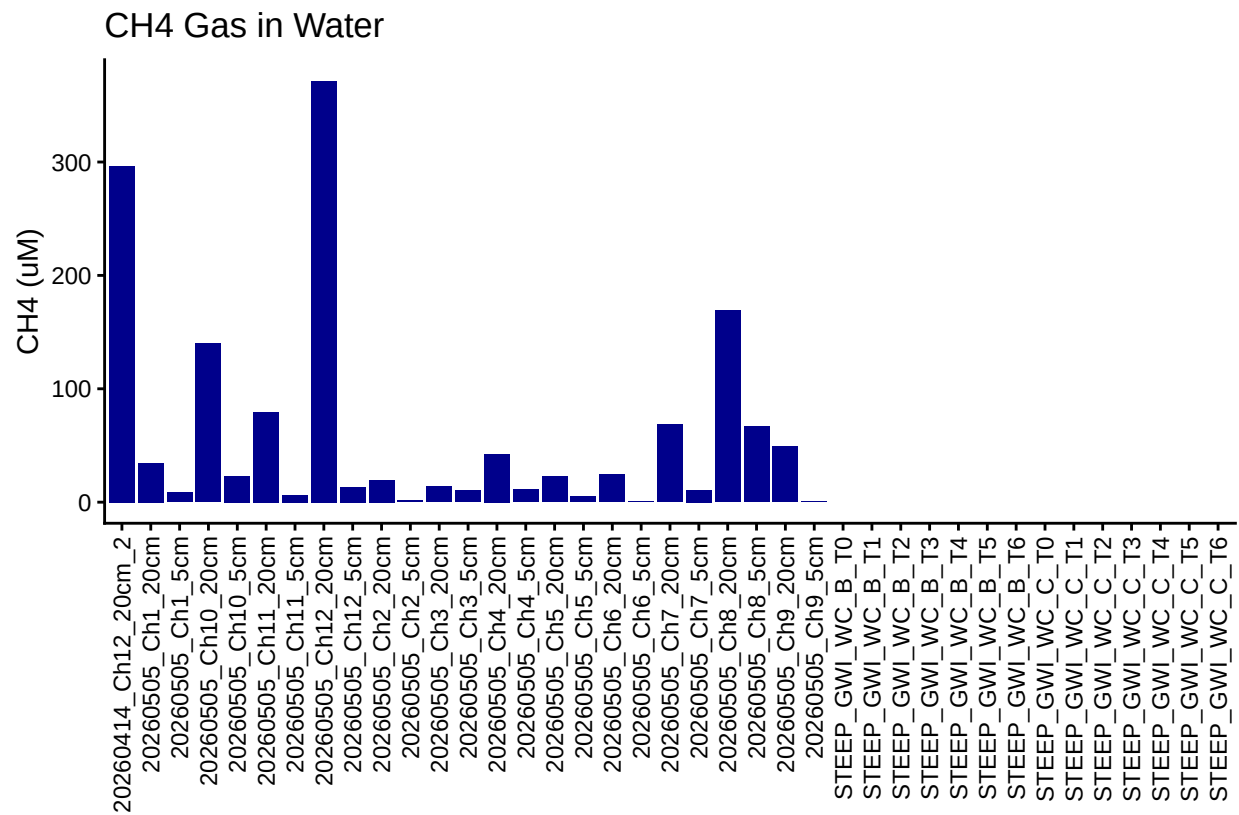
5 Dilution correct samples and Check Range



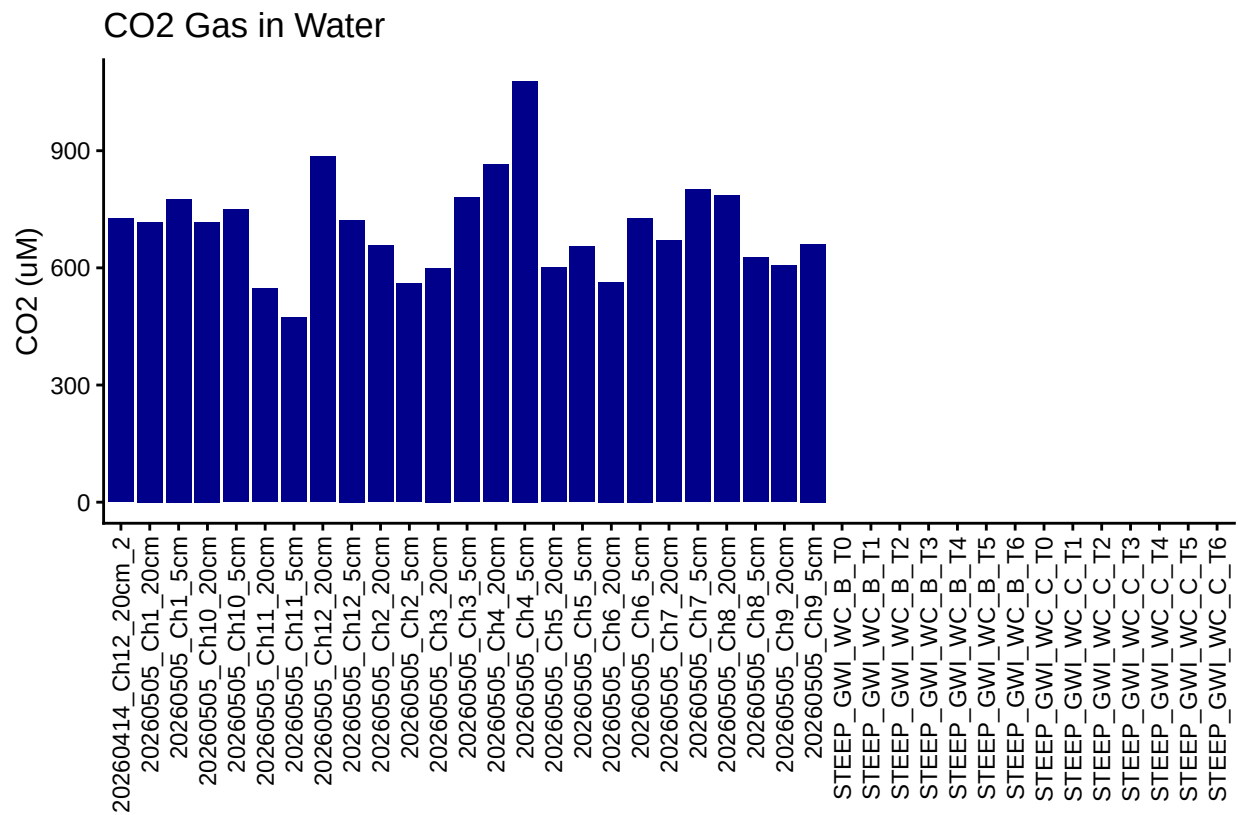


6 Calculate gas in water

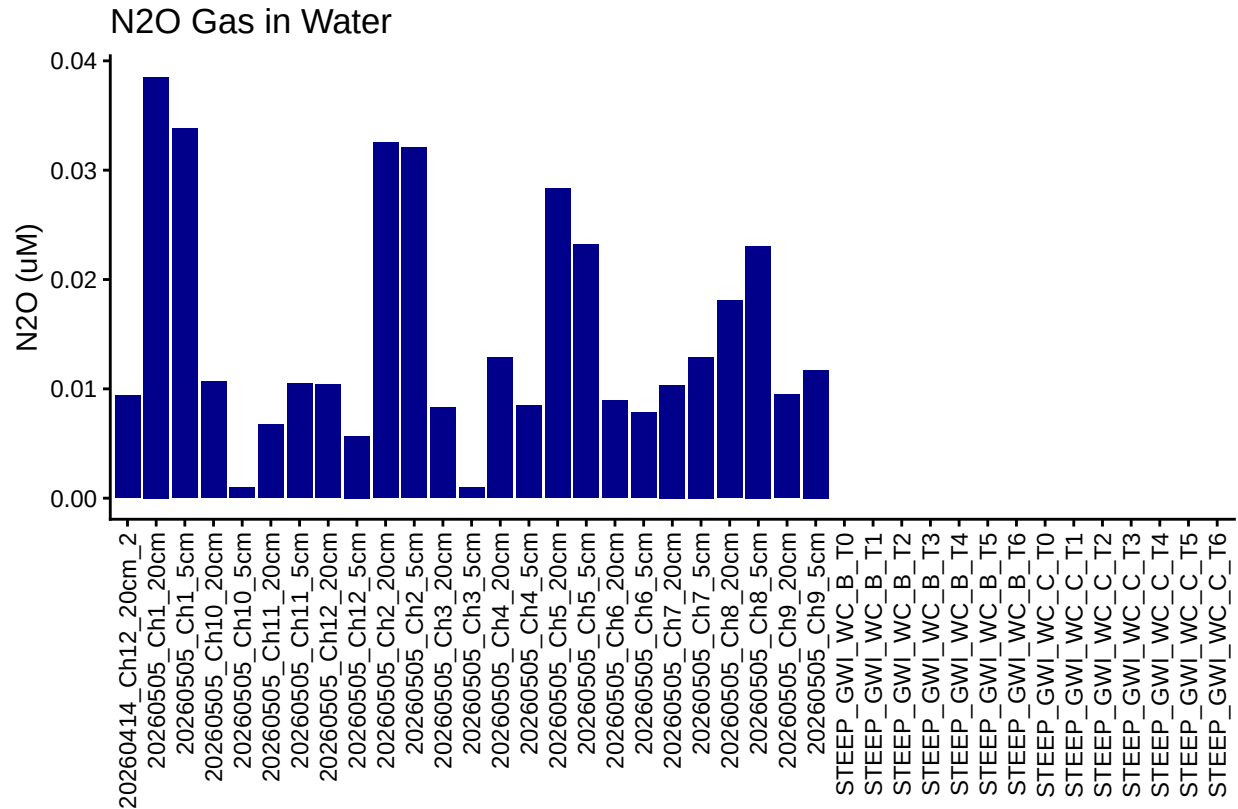
```
## Warning: Removed 14 rows containing missing values or values outside the scale range
## ('geom_bar()').
```



```
## Warning: Removed 14 rows containing missing values or values outside the scale range
## ('geom_bar()').
```



```
## Warning: Removed 14 rows containing missing values or values outside the scale range
## ('geom_bar()').
```



7 Merge samples with Metadata & check all samples are present

```
## Merge Samples with Metadata
```

```
## Some sample IDs are missing from data.
```

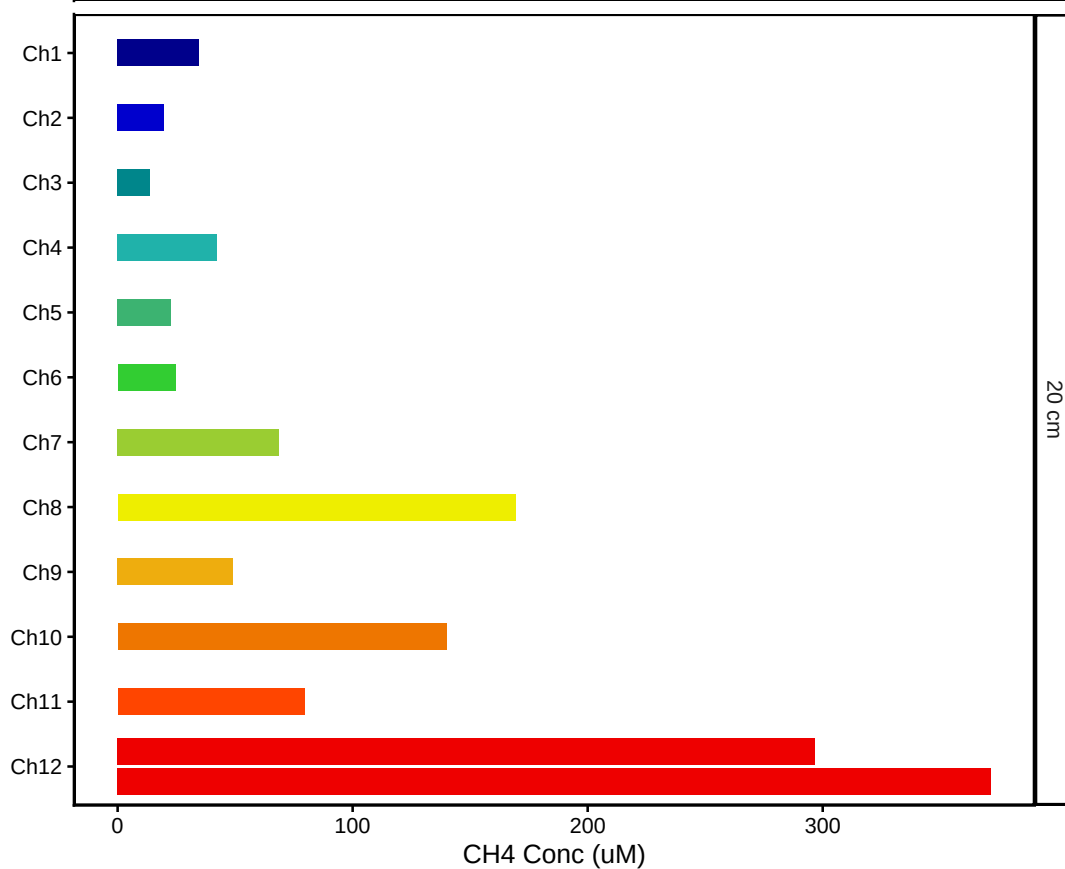
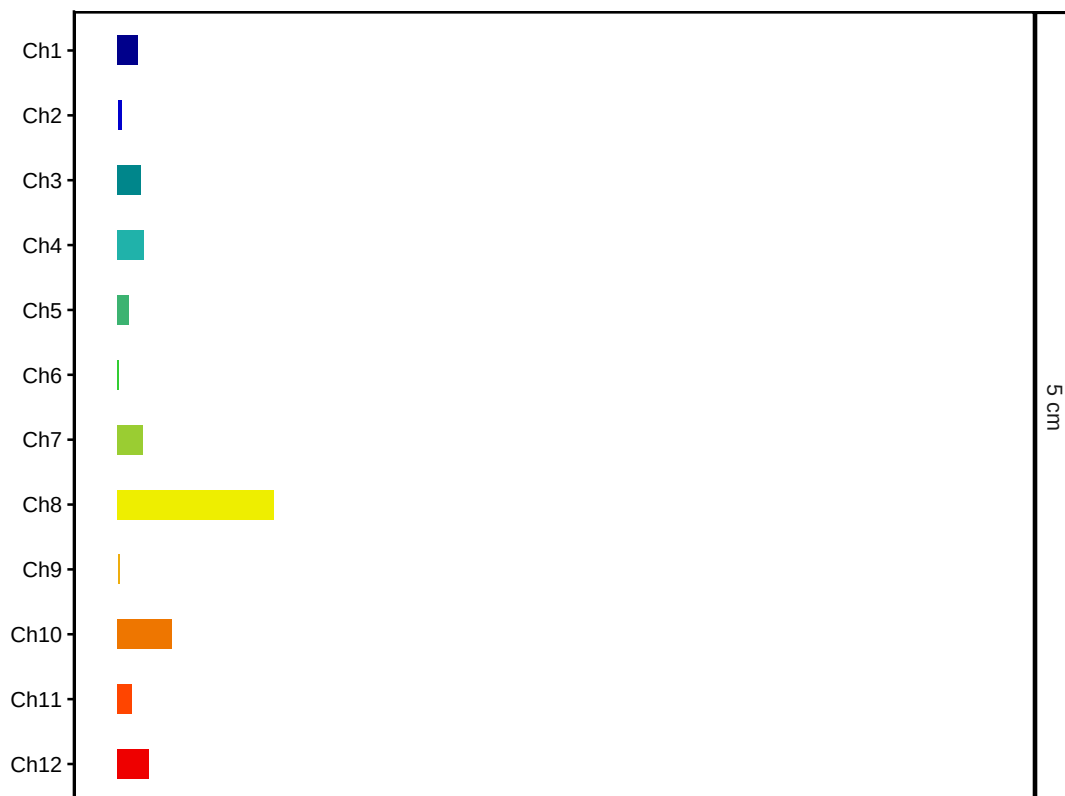
```
## [1] "Ch1-40" "Ch1-80" "Ch2-40" "Ch2-80" "Ch3-40" "Ch3-80" "Ch4-40"
## [8] "Ch4-80" "Ch5-40" "Ch5-80" "Ch6-40" "Ch6-80" "Ch7-40" "Ch7-80"
## [15] "Ch8-40" "Ch8-80" "Ch9-40" "Ch9-80" "Ch10-40" "Ch10-80" "Ch11-40"
## [22] "Ch11-80" "Ch12-40" "Ch12-80"
```

7.1 Organize Data

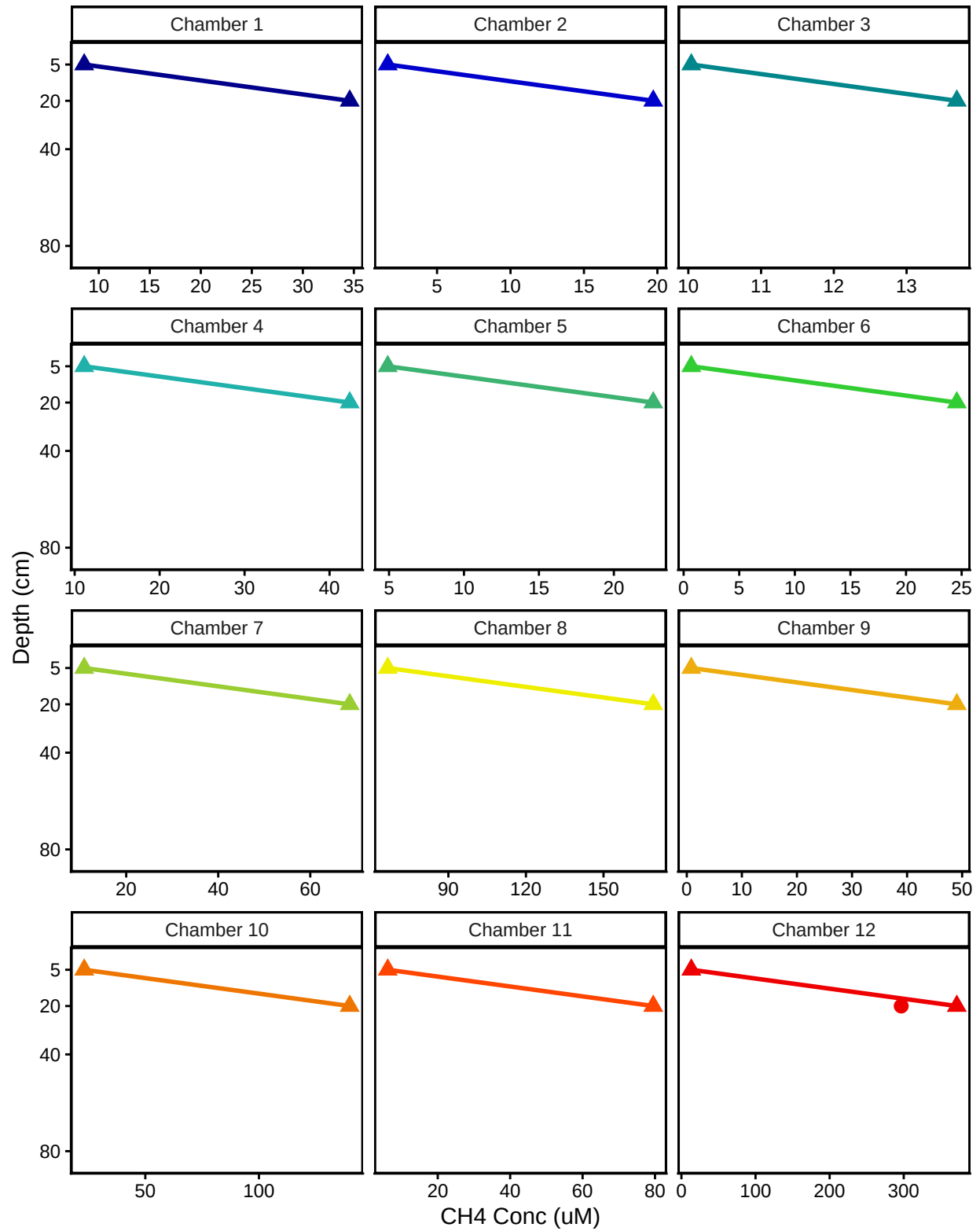
7.2 Visualize Data CH4

```
## Visualize Data
```

GENX: Porewater CH4



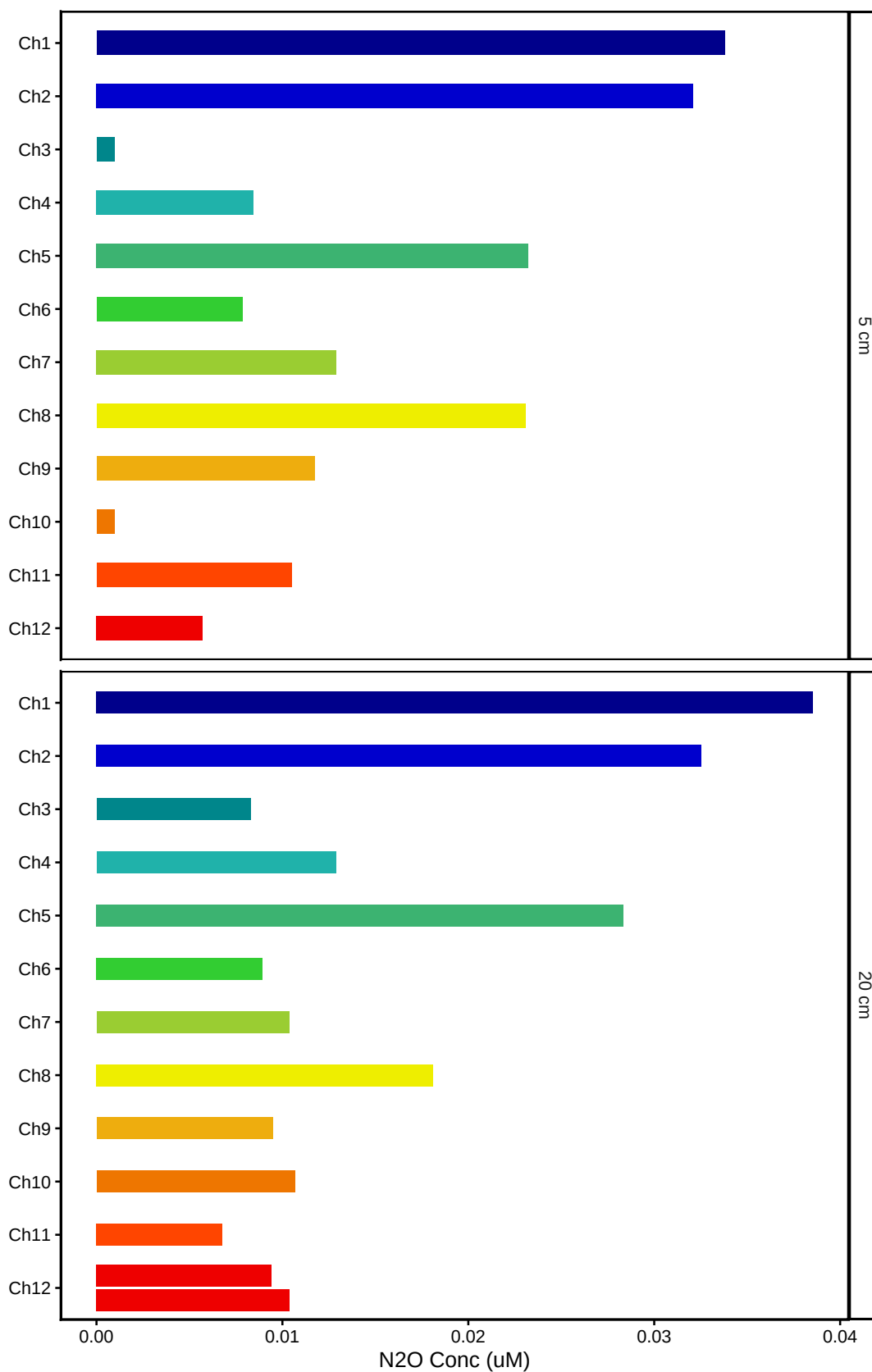
GENX: Porewater CH4 by Chamber



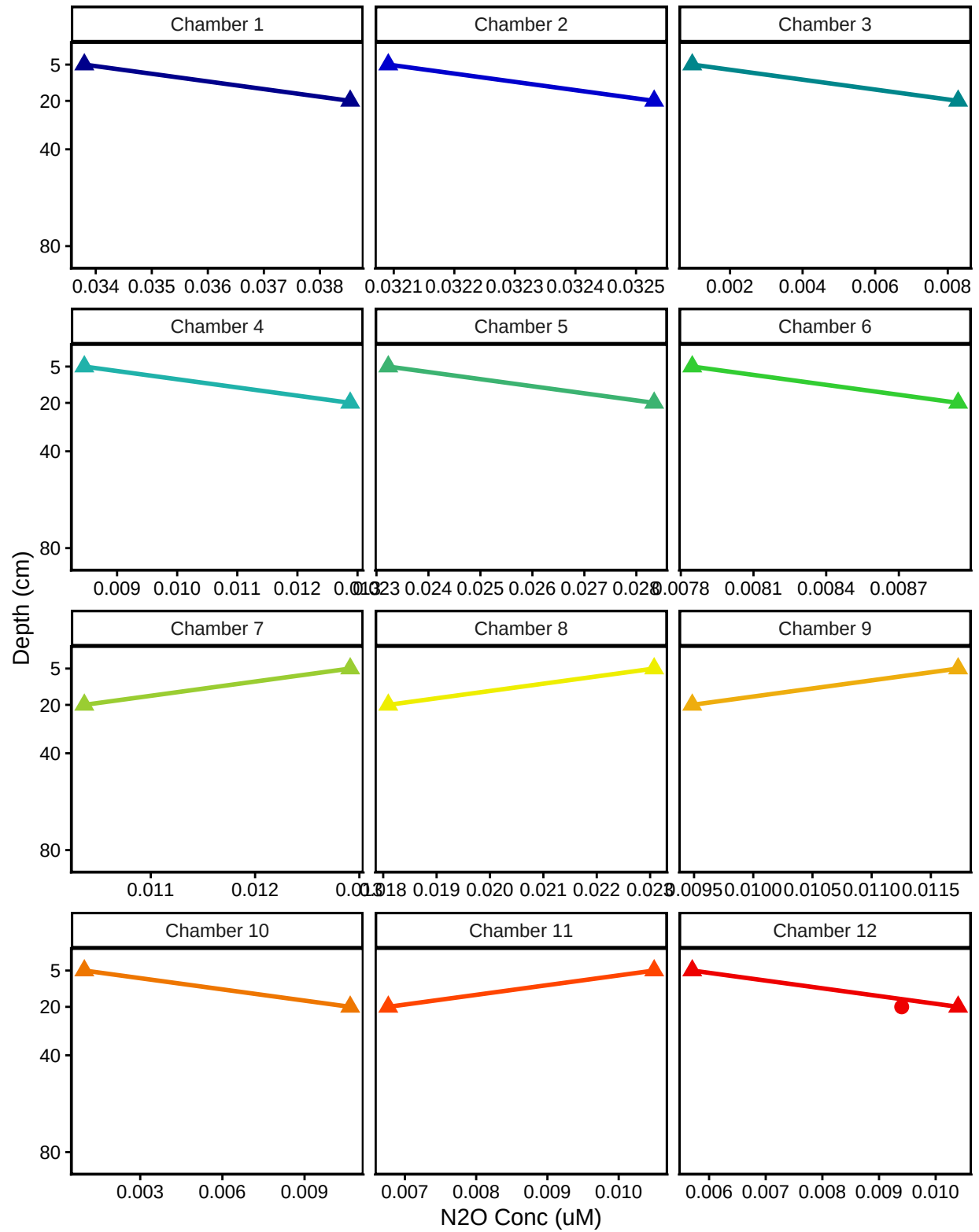
##Visualize Data N2O

```
## Visualize Data
```

GENX: Porewater N2O



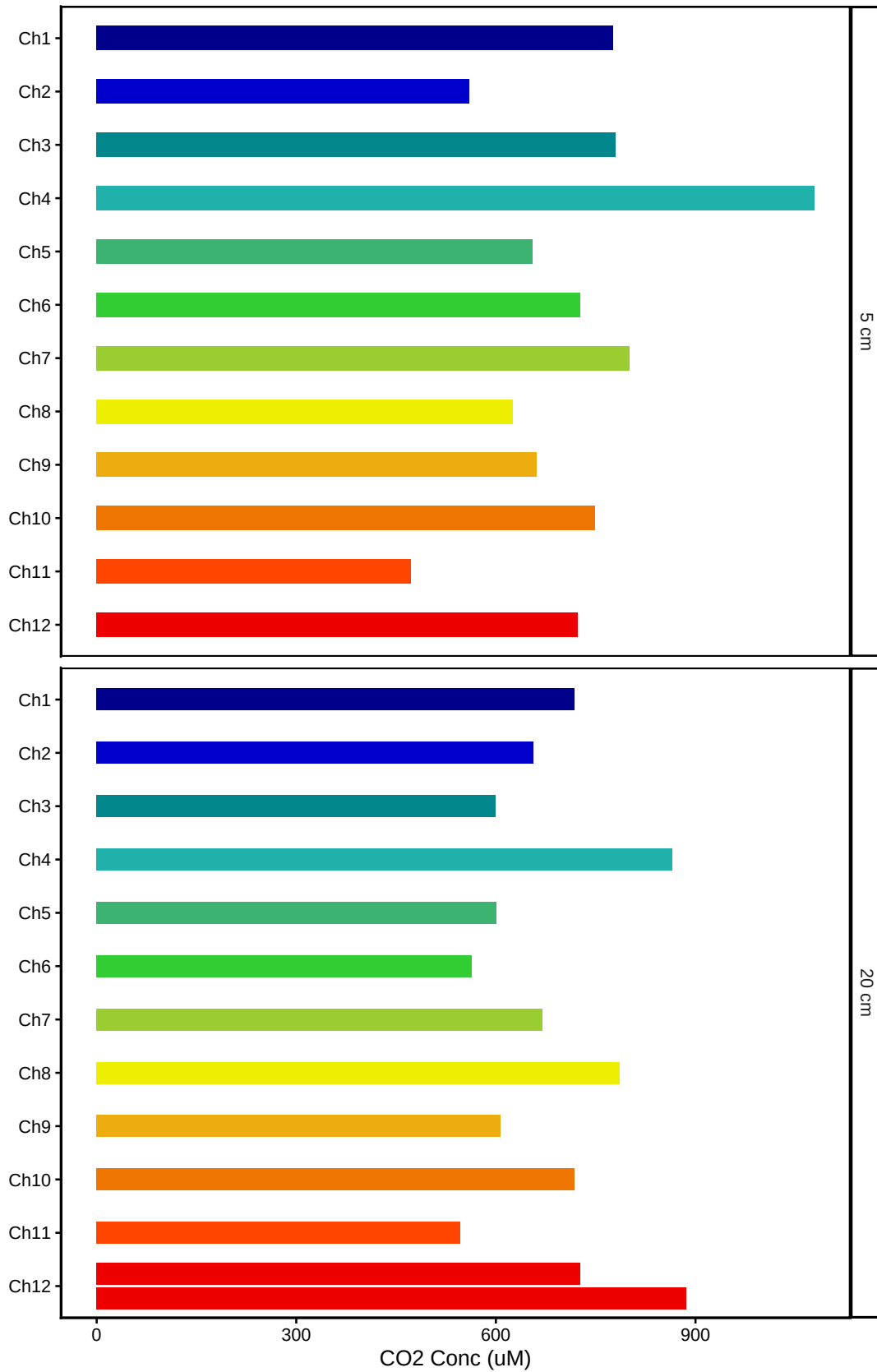
GENX: Porewater N2O by Chamber



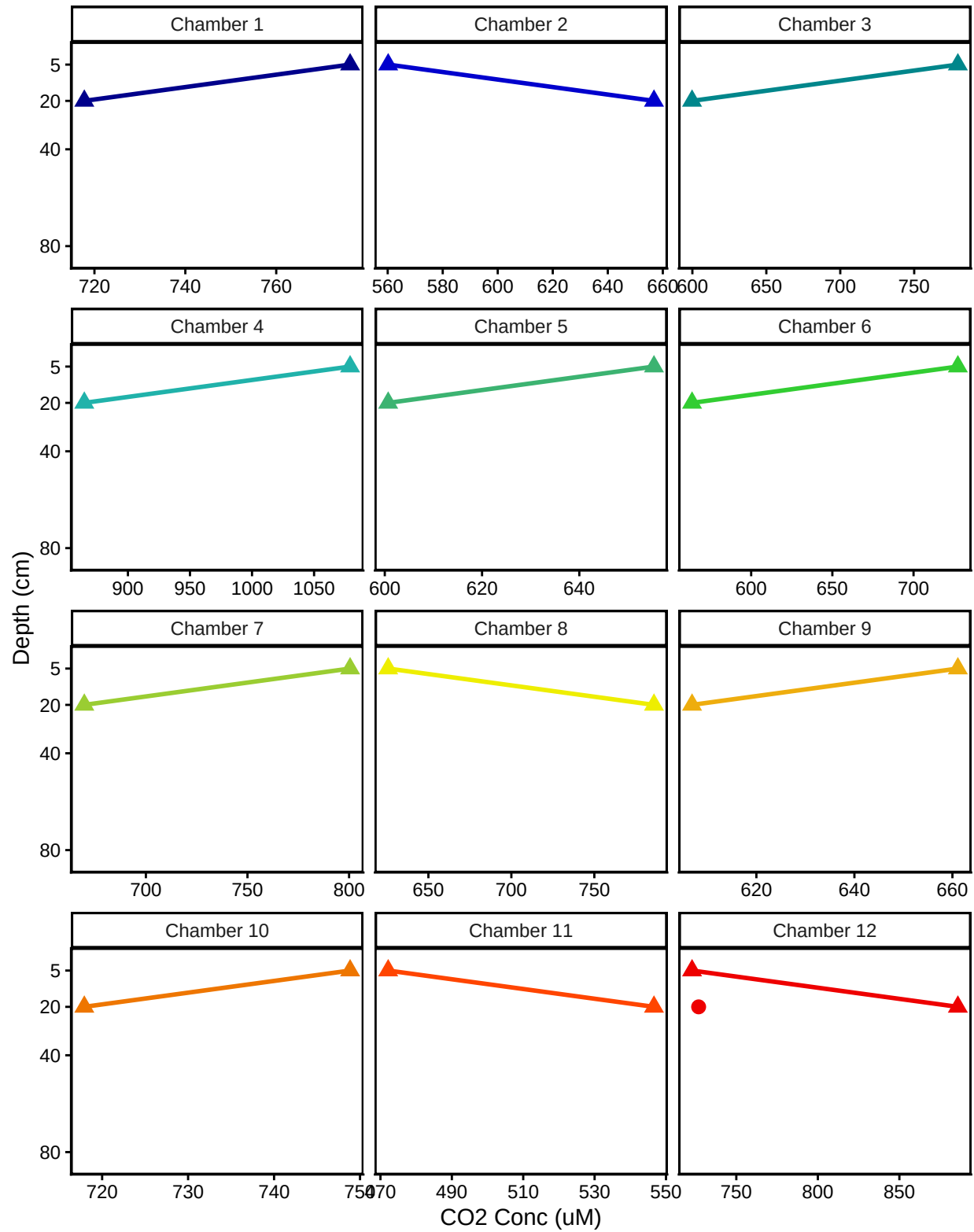
##Visualize Data CO2

```
## Visualize Data
```

GENX: Porewater CO2



GENX: Porewater CO2 by Chamber



##Export Data

#end